

Research dossier

Deadline constrained AI and quantum optimisation for Formula 1 strategy

Version 3.1 | Research design and execution specification

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This corrected revision supersedes version 3.0 and retains the redesign of version 2. It proposes a study; it reports no new experimental results. Implementation, protocol freeze, physical quantum execution, manuscript writing and journal selection remain subsequent stages.

Decision. Proceed with a staged, zero additional spending research project. The strongest defensible direction is an auditable study of when a learned quantum candidate generator can contribute to a two-car pit decision before a deadline, and when the available classical solution or model uncertainty leaves it no useful role.

The immediate deliverable is a reproducible motorsport decision benchmark, a learned parameter-selection component, carefully controlled circuit comparisons, and an empirical map of their limits. Demonstrating benefit in actual Formula 1 operations requires additional evidence beyond a simulator. Neither this dossier nor a successful hardware run establishes that benefit.

The design changes materially. Small one-hot formulations are treated as compiler and sampling tests, not evidence of computational difficulty. Historical timing data becomes an optional, permission-gated validation track. Local experiments carry the statistical workload. Physical hardware has a narrowly defined validation role within the reported 540-second balance.

Absolute novelty, future quantum advantage, publication acceptance and recruitment by a team cannot be guaranteed. The study is designed to earn a specific, bounded contribution through comparisons and reproducible evidence, including evidence that quantum execution is presently unnecessary.

What remains non-negotiable

AI must perform an evaluated runtime function. The quantum component must be physically tested if the preceding scientific and budget gates pass. The decision problem must retain meaningful motorsport mechanisms. No study-specific paid services are assumed. The completed earlier study and its evidence remain unchanged.

This is independent research. No association with, endorsement by, or access to a Formula 1 team is represented.

Corrections in version 3.1

This review found implementation and inference gaps that needed correction before coding. No experiment has been conducted or reinterpreted. The changes below form part of the specification and take precedence over version 3.0.

Issue found	Correction
Nominal deadline could exceed pit-entry opportunity	Effective deadlines now include action expiry, state freshness and a commitment margin.
Development data could fit, tune and calibrate the same model	Training, tuning and calibration have separate block assignments and frozen roles.
Readiness score called a confidence estimate without a coverage argument	It is now an empirical prediction score, with separate held-out coverage checks and no guaranteed improvement claim.
Circuit matrix assumed an optional third circuit on every case	C0 and C1 define the core; C2 is a separately registered conditional local study.
Circuit comparisons and parameter-policy ablations were incompletely counted	The matrix states which counts are per parameter policy and which experiments remain conditional.
Dropping repetitions did not fix an individual job's insufficient cap	Pilot-based reservations and two alternative complete hardware schedules replace the ambiguous fallback.
Quantum penalty and parameter-scaling conditions were incomplete	The feasibility assumptions, a sufficient penalty bound and normalisation rules are explicit.
Repeated stochastic decisions and simulation uncertainty were underspecified	Policy seeds, paired blocks and a single primary resampling procedure are specified.
Small exact cases could trigger an uninformative large superiority study	A pre-test headroom gate prevents powering a mathematically unavailable improvement.
Generic learned solver allocation had insufficient prior-art coverage	Algorithm-selection precedent is now cited directly.

What this correction does not establish

The research design still needs its implementation pilot, frozen simulator parameters and source permissions for any historical extension. Those are declared future tasks, not completed checks. Neither this review nor the proposed run counts establishes absolute novelty, operational F1 benefit or future quantum superiority.

The main restrictions remain unchanged: no additional spending, the reported 540-second QPU allowance, a protected reserve, immutable evidence, and the copy-paste Cursor workflow.

1 How to use this specification

The project proceeds in three distinct stages: finalise this research design; implement and collect evidence through Cursor; then write the article and select a suitable journal. A plan is not a result, and a generated configuration is not a frozen protocol until its unresolved feasibility choices have been recorded.

Reading map

Sections	Purpose
2 to 4	Corrections, related work and the proposed contribution
5 to 10	Decision domain, data, simulator, formulation and exact references
11 to 15	AI, circuits, acceptance and deadlines
16 to 19	Comparators, experiment counts and statistical analysis
20 to 22	Physical hardware campaign and conditional readiness analysis
23 to 27	Cursor workflow, evidence, implementation and claim limits
28 to 30	Sources and the protocol freeze record

Fixed constraints

The user is a non-coder and has no assumed circuit-design expertise. ChatGPT prepares complete, scoped prompts; the user copies them into Cursor. Cursor must eventually translate “do a run” into one clearly identified, resumable unit of the frozen plan. It must save the raw evidence and report actual completion. Pushing to GitHub remains a separate requested action.

Only existing paid subscriptions may be used. Core work uses local computation and free software. The hardware allowance is the user's reported nine minutes, provisionally 540 seconds, to be checked against the provider account before any job. It is not an authenticated balance in this document.

The closed project is not reopened. Its manuscript, 20 historical GHZ observations and other decision-study evidence are not reused as new F1 results. A new isolated repository is preferred. If the existing repository is retained, use a separate top-level directory and enforce an allowlist of writable paths.

Meaning of the labels

Verified refers to a checked source or a completed implementation check with evidence.

Proposed denotes a research choice in this dossier. **Conditional** means the choice depends on a stated gate. **Measured** may be used only after a run produces its raw record. These distinctions apply to every report, table and later manuscript.

Approving the dossier is not a command to execute experiments or publish results. Subsequent implementation and run instructions determine the authorised scope.

2 Corrections to version 2

Version 2 contains useful priorities: deadline accounting, fallback decisions, strong classical references, independent counterfactual evaluation, learned parameter transfer, limited hardware use and immutable evidence. Those priorities are retained. Several supporting claims require replacement.

Version 2 issue	Required correction
24 binary variables imply 2^{24} relevant choices	Exactly one selected action gives only 24 legal choices; a stay option may add one. Count legal plans, not unused bit strings.
Published timings settle unmatched current comparisons	Timings concern particular models, machines and objectives. Benchmark equivalent problems; do not infer a universal impossibility.
A predictor must beat an optimised ceiling	Compare with a best-found offline reference using non-inferiority, and with simpler runtime parameter policies using superiority.
No predictor means no deadline-feasible quantum arm	Fixed angles and lookup transfer also avoid an online optimisation loop. They are required baselines.
Identical candidate pools across solvers	Use identical information, objectives and budgets. Candidate generation must be allowed to differ.
Same seed guarantees effective CRN	Couple random quantities by event identity; measure the covariance and variance reduction.
Simulator model error is inherited from citations	Validate this implementation and corpus. Sampling intervals do not cover structural model error.
Historical longest stint is a safety limit	Treat it as a prior or modelling assumption, not a physical or regulatory maximum.
Deadline admissibility is a new general theoretical result	Deadline-aware metareasoning has precedent. Establish a narrower empirical or methodological contribution.
Replication and sample-size justification are unspecified	Match sample size to effect, variance, dependence and precision. Hardware evidence remains small.

The revised design also removes an implicit guarantee that improved quantum hardware will eventually win. A faster processor cannot create value where encoding has already revealed the optimum, where model error dominates the action gap, or where a stronger classical method solves the same task first.

The empirical question is therefore conditional: under which explicitly measured combinations of decision difficulty, uncertainty, circuit quality and latency is a quantum candidate useful?

3 Related work and novelty boundaries

The literature check establishes nearby precedents, not an exhaustive proof of originality. Search again before protocol freeze and manuscript submission, recording queries, dates, versions and full texts actually inspected.

Race strategy optimisation. Carrasco Heine and Thraves develop deterministic and stochastic dynamic programming for pit strategies [1]. Their model is a baseline source, not proof that every traffic-coupled team problem is solved. The accessible implementation provides a route to validating a restricted DP reference [2].

Direct F1 QUBO precedent. Sophie Kolstee's 2026 Leiden bachelor's thesis, previously unverified in version 2, is available [3]. It studies QUBO formulations using simulated and quantum-inspired annealing on synthetic instances. It does not report physical QPU experiments. A first-F1-QUBO claim is unavailable.

Learned racing decisions. Thomas and colleagues study reinforcement-learning pit strategy with a proprietary Mercedes simulator [4]. Fieni and colleagues study learned multi-agent racing strategies [5]. Their existence rules out a generic first-AI-strategist claim; their inaccessible components must not become a hidden dependency.

The nearest systems precedent. Santillana's Pitwall preprint describes calibrated Monte Carlo strategy evaluation, common random numbers, verified briefings and author-reported live deployments [6]. Those deployment claims are attributed to the preprint, not independently verified here. A generic grounded, deadline-aware strategy pipeline is not new.

Quantum methods. QAOA, alternating-operator constructions and warm starts are established [7-9]. Graph-based and cross-problem parameter transfer also precede this proposal [10-12]. Gupta and Hassan explicitly develop guarded XY moves and collective exchanges for rostering [13]. Adapting such moves to pit decisions is not inventing the general technique.

Simulation and decision theory. Learned algorithm selection is established [27]; it is not made novel by renaming it an allocator. Common random numbers and metareasoning under limited resources have prior theory [14-15]. Galbraith's information-processing perspective may motivate organisational relevance [16], but "late computation has no value" is not a new general theorem: late results can still support learning or later decisions.

Candidate contribution

Test a margin-aware allocation policy that links learned quantum sampling, uncertainty in the decision model and the complete decision deadline. Publish the cases in which it declines quantum execution, as well as those in which it accepts it. Its originality must be assessed against algorithm selection, metareasoning, racing strategy and quantum benchmarking together.

The contribution is provisional until the difference from these methods is implemented, tested and shown to matter.

4 Questions and proposed contributions

Primary question. Does the proposed allocation policy improve the quality of a deadline-bounded team decision over the strongest included classical policy selected on tuning data, under the same information and end-to-end resource accounting?

Supporting questions. Does learned parameter selection outperform fixed and nearest-neighbour transfer? Does a structured circuit yield more useful feasible candidates per unit of execution cost? How much of any apparent optimisation improvement survives independent race simulation? Which delays, errors or modelling assumptions prevent useful quantum participation?

Three contributions to attempt

A decision benchmark. Release generated scenario definitions, causal checkpoint states, exact small-instance references, solver interfaces and an auditable deadline protocol. A benchmark is valuable only if it tests meaningful failure modes and can be reused; “open source” alone is not novelty.

A bounded allocation method. Combine an incumbent, a resolvable improvement margin, a learned estimate of candidate success, and a measured or explicitly modelled latency budget. Dispatch only when there is credible room for improvement. Evaluate against always-classical, always-quantum and simple threshold allocation.

A hardware-linked boundary study. Measure how ideal, noisy and physical circuits change useful-candidate yield, then identify conditional latency and sample-quality requirements. Report these as requirements within a defined benchmark, not universal specifications for a future F1 computer.

Proposed hypotheses to freeze

H1 estimates the paired difference in simulated team loss at the nominal 30-second deadline, shortened where commitment cutoffs require. H2 compares learned and simple parameter policies on optimal-sample probability and regret. H3 compares C1 with C0 on raw feasibility and optimal-sample probability. H2 and H3 are exploratory mechanism analyses; hardware is descriptive validation. The H1 superiority test proceeds only if the pre-freeze headroom gate permits it.

The planned minimum worthwhile H1 effect is 0.02 on the normalised team-rank loss defined in section 9. This is a research threshold, not an elicited team preference. Show the corresponding unnormalised positions and sensitivity at 0.01 and 0.05. Freeze the threshold before test outcomes are viewed.

Failure remains informative but not automatically publishable

If AI adds no measurable value, remove the AI-performance claim. If all decision instances have zero optimisation headroom, report that fact and restrict quantum claims to sampling and integration. If no substantive difference from prior work remains, redesign before spending hardware time. A negative result needs a meaningful question and reliable measurement.

5 Decision domain and scope

The core domain is a dry-weather, two-car team choosing whether to pit now, delay by one or two completed laps, or continue with a frozen feasible continuation policy after a safety-car or virtual-safety-car checkpoint. The choice includes a tyre compound and an available set. The simulator carries the consequences to race end.

Use a synthetic 20-car field for the core benchmark. This is a declared model setting, not a claim about the grid size or rules of every Formula 1 season. Parameterise field size, points schedules and sporting rules rather than hard-coding “19 rivals.” Any historical extension pins the applicable season and event documents [17].

State available at the checkpoint

Record lap, race-clock position and each car's pit-entry commitment cutoff; safety regime and information known about its duration; positions and gaps; compound and tyre age; fuel estimate and uncertainty; tyre-set inventory; pit-lane status; team service state; completed compound obligations; and the remaining horizon.

Every field carries a source, unit, availability time and known, inferred or assumed status. Decisions use the information available at the checkpoint. They do not see the sampled future safety-car duration, a rival's eventual pit lap or the current lap's final time.

Coupling that is worth modelling

Two team cars may interfere through double stacking, rejoin traffic, service order and position trade-offs. Rivals can react under a specified policy. Fuel, tyre wear and traffic affect the continuation. These mechanisms create decision relevance; merely adding binary variables does not.

Double stacking is normally a delay mechanism, not a universal prohibition. Model the shared crew and the second car's waiting time. A “never double stack” rule is an optional declared team policy, not an FIA requirement. Do not invent such a prohibition to create a difficult constraint graph.

Initial exclusions

Wet transitions, red flags, sprint-specific rules, tyre damage and detailed energy deployment are excluded from the primary experiment. A checkpoint requiring an unsupported mechanism is rejected with a recorded reason. Later stress tests may add one mechanism at a time, with separate results.

The fallback is a feasible team plan revalidated at commitment, including a legal continuation. “Stay out” is an explicit candidate when feasible, not an unconditional substitute for compliance with the remaining race rules.

6 Data provenance and real-world validation

Core data are generated. Use independently specified, fictional drivers, teams and track archetypes. Publish the generator and its distributions. Do not present synthetic checkpoints as historical races or evidence of deployment.

Formula 1's published guidelines restrict AI use of its protected rights without an express licence [18]. These are the rights holder's stated conditions, not a determination that every historical fact or independent dataset is legally unavailable. Document the lawful basis and applicable source terms for the intended use. FastF1's MIT software licence does not itself license underlying timing records [19].

Accordingly, no protected timing feed is required for the core study. Before any historical-data track, record the source's terms and the permission covering the intended analysis, learning and publication. A software licence or a downloadable file alone is insufficient. No paid licence is assumed.

Open software and input files are separate

The TUM race-simulation repository is a candidate implementation base under LGPL-3.0 [20]. Audit bundled configurations and data separately before reuse. Preserve applicable notices and modifications. If a required input lacks clear provenance, replace it with a documented generated input; do not silently treat it as open data.

Optional historical validation

Proceed only after suitable permission or a clearly applicable data licence is documented. A candidate chronological split is 2018-2021 for training, 2022 for tuning, 2023 for calibration and 2024-2025 for testing, subject to actual lawful coverage. This split is proposed, not an existing corpus.

Aim for at least 30 eligible held-out races across at least 10 circuits and both SC and VSC conditions. A census must establish whether that coverage exists. Additional checkpoints within the same race do not substitute for independent races. If coverage is smaller, narrow the claim and calculate precision from the realised design.

Reconstructed historical availability is not live receipt-time evidence. Finalised lap archives can leak later information. Tyre-set inventory and team intentions may be unavailable and must remain explicit assumptions.

The route to actual F1 value

Synthetic evidence can establish an algorithm's behaviour and expose integration requirements. Historical validation can test plausibility and calibration. A team's prospective shadow trial would be stronger evidence of operational value. These are distinct levels; only the level actually completed may appear in the article's claims.

This dossier authorises no contact with a team or data owner. The method and evidence package can be prepared before any later outreach is requested.

7 Generated corpus and generalisation

Use eight families from three factors: low or high pit loss; near-linear or nonlinear tyre degradation; and sparse or dense traffic. Each independent parameter block generates eight separate episodes with causal checkpoints, four conditioned on SC and four on VSC. Advance valid states before applying the declared regime at a prescribed checkpoint. This balanced design does not estimate natural SC/VSC frequencies.

Split	Independent blocks	Checkpoints
Training	120, 15 per family	960
Tuning	16, 2 per family	128
Calibration	24, 3 per family	192
Primary test floor	80, 10 per family	640
Separate shift panel	40	320

Training, tuning, calibration and primary test total 1,920 checkpoints at the floor; including the shift panel gives 2,240. Section 18 may raise the test count to 160 blocks before test access. Round increases to multiples of eight to preserve balance. Worlds and policy seeds do not add independent blocks.

Roles and provenance

Fit donor models and dispatch models on training blocks only. Select architectures, hyperparameters, circuit depth and the included classical comparator on tuning blocks. Freeze those choices before opening calibration blocks, which estimate score corrections, variance and replication counts. They do not train or retune the models. If calibration causes a model redesign, retire those blocks to development and generate a fresh calibration set before test access.

Store block seed, generator version, parameters, family and split. Distinct keyed streams control episodes, coefficients, fitting, online scoring and final evaluation. All descendants remain in the parent split. Selection by favourable solver outcomes is forbidden.

Starting ranges are declared assumptions: 20 cars; 12-45 remaining laps; green laps of 75-110 seconds; green pit loss of 18-28 seconds; and explicit tyre, traffic and restart parameters. Audit and freeze the full distributions during the pilot. These ranges are not a calibrated estimate of F1 conditions.

Panels and action menus

The primary test samples new blocks within the eight families. The shift panel changes predeclared ranges or combinations and remains separate. It does not establish generalisation to actual circuits or seasons.

Select 120 small quantum test cases from at least 60 test blocks and 12 hardware cases from distinct blocks by a frozen stratified hash. Freeze reduced action menus before solver outcomes and use them for all comparators. Report excluded actions. Restricted hardware cases do not test the complete menu.

8 Simulator and counterfactual validity

A changed pit decision changes the continuation. Static historical replay cannot reveal the unobserved alternative. The outcome in this study is conditional on a simulator, its policies and its uncertainty assumptions.

The preferred starting point is a pinned adapter around TUM's full race simulator, subject to compatibility and checkpoint-resume tests [20]. Its basic free-track variant is useful for analytical checks but cannot validate traffic interactions. If the full simulator cannot support the required checkpoint semantics, implement a restricted independent model and narrow the claims; do not imply that repository reuse supplies missing validation.

Required mechanism checks

Check tyre-age and fuel updates, event ordering, pit entry and exit, shared-crew service, rejoin ordering, SC bunching, VSC pace control without equivalent bunching, restart transitions, compound obligations and end-of-race classification. An SC represented only by multiplying every lap time is insufficient for the primary SC claim.

Resume a saved checkpoint and compare its continuation with an uninterrupted run under identical exogenous draws. Verify conservation of race-clock ordering, nonnegative service and travel times, legal tyre transitions and a consistent treatment of retirement. Compare simple no-traffic cases against an independently coded analytical or DP model.

Avoid a self-confirming evaluator

The objective compiler and final evaluator must not be the same function under different names. Build coefficients from a restricted model; score decisions with additional traffic and stochastic mechanisms. Validate both independently in cases with known answers.

The final evaluator still shares assumptions with the generator. Address this by predeclared sensitivity models: changed degradation curvature, service variability, restart compression and rival responsiveness. Agreement between related models is robustness evidence, not empirical validation of real races.

Rival response

The core rival policy is frozen before test. It may recompute a feasible remaining strategy from its own observable state. Use a simpler non-reactive policy and a more responsive policy as sensitivity analyses. Do not feed actual future rival actions into an online comparison.

Monte Carlo confidence intervals describe sampling variation conditional on these choices. Model discrepancy is addressed by calibration where lawful observations exist, cross-model checks and sensitivity analysis, and remains a limitation where it cannot be measured.

9 Objective and QUBO specification

For car c , let $A(c)$ be its legal action plans. Each plan contains an immediate pit choice or continuation plus a specified downstream policy. Binary $x(c,a)$ selects plan a , with exactly one plan selected per car.

Proxy objective. Minimise the sum of per-plan costs and pairwise interaction costs. Per-plan cost covers predicted remaining time under the compiler model. Pair terms represent additional shared-crew and traffic effects. Do not add a separate degradation or traffic penalty if it is already included in predicted time.

$$f(x) = \sum a(c,a)x(c,a) + \sum b(a,b)x(1,a)x(2,b).$$

Before setting penalties, subtract each car's minimum unary cost from its unary coefficients and retain the offset as a constant. One-hot legality makes this exact on legal plans. It prevents a large shared race-time baseline from inflating penalties. Recompute bounds after centring and restore constants for physical-unit reporting.

Any risk term must have explicit units and a linear or quadratic representation, or a separately verified reduction with auxiliary variables. Do not assume an arbitrary tail-risk functional is a QUBO. Choose weights on tuning data and show sensitivity including zero. Hard legality and risk preference are distinct.

QUBO. Add one-hot penalties $M[\sum x(c,a) - 1]^2$ and any genuine additional hard-constraint penalties. Remove individually illegal actions before encoding. Document upper-triangular coefficient conventions, constants and the mapping $x = (1 - Z)/2$; verify energies and decoding exhaustively on small cases.

Let B be the sum of absolute unpenalised coefficients, excluding the constant. Their objective range over all bit strings is at most B . Require $P = 0$ on every feasible state, $P \geq v_{\min} > 0$ otherwise, and at least one feasible state. Then $M > B/v_{\min}$ suffices for $f + MP$: an infeasible penalty exceeds any possible unpenalised gain. Choose a strictly positive numerical margin and verify the encoded constraints. Version 3.0 used a looser bound but omitted the feasible-zero condition.

Downstream outcome

For field size F and the two cars' classified ranks r_1 and r_2 , define normalised team-rank loss $L = (r_1 + r_2 - 2) / [2(F - 1)]$. Lower is better. Retirements follow a frozen classification rule. Report expected loss, individual ranks, team points under the chosen schedule, and a downside measure separately.

The primary comparison is classical-policy loss minus proposed-policy loss, so positive values favour the proposed policy. This endpoint measures the modelled team objective. It is not automatically equivalent to championship utility or a team's actual preferences.

A smaller f does not guarantee smaller expected L . Report ranking disagreement and decision regret between the proxy and the independent evaluator. This distinction is central to interpreting any solver result.

10 Exact references and honest scaling

With K actions for one car and one-hot selection, there are K legal plans. With two cars, there are at most $K_1 \times K_2$ pairs. Six actions per car use 12 qubits but produce only 36 legal pairs. Enumerating millions of invalid bit strings would manufacture a classical disadvantage.

If the compiler computes a complete table of every pair's objective, its minimum can be obtained by a cheap scan of that table. In that case quantum execution tests encoding and sampling, not a need for optimisation. Include the table construction and minimum search in the classical comparison.

Three distinct references

Exact proxy optimum. Enumerate all legal choices in the small action set. This certifies the minimum of f , not the best real race decision.

Finite-simulation reference. Evaluate every legal pair using a fixed offline Monte Carlo bank on a designated subset. The best estimated action is an offline reference with sampling uncertainty, not an omniscient true optimum. Use an independent bank to estimate the selected reference's performance.

Large-instance bound. For extensions, use a free integer solver with valid bounds and recorded termination gaps. A DP reference is exact only for the assumptions its state and transition structure represent [1-2].

Implement a linearised binary model independently of the QUBO builder. For a product $z = xy$, enforce $z \leq x$, $z \leq y$ and $z \geq x + y - 1$. If an integer solver requires coefficient scaling, quantify rounding error and ties. Check both formulations against enumeration before using the solver as a reference.

Conditional scaling extension

Treat a multi-stop horizon with team interactions as a separate model extension. It needs its own variables, transition constraints, QUBO reduction, bounds and frozen protocol after the small model passes validation. Publish the mapping from every extra variable to a decision. Keep problem semantics matched across solvers; report preprocessing, memory and wall time.

A conditional stress panel targets 1,200 instances across horizon, action count and interaction density; it is not authorised by the current single-choice formulation alone. It is classical-first. Small feasible members can also be simulated quantum mechanically; larger members are resource estimates or classical benchmarks unless actually executed.

No crossover is guaranteed. If classical methods solve every meaningful instance well within the deadline, the correct result is no observed optimisation headroom. A tiny hardware experiment cannot establish how a larger quantum algorithm will scale. Check this on development cases before launching a large campaign: do not power a superiority test for an improvement that the formulation makes impossible. Before the full test campaign, either freeze a meaningful larger model or record H1 as descriptive boundary evidence and omit its superiority test. If no nontrivial scientific question survives, stop the campaign and redesign.

11 The AI component

The runtime learned component selects a QAOA donor from an offline angle bank. A donor ranker avoids averaging incompatible periodic angle vectors. It is distinct from the optional learned dispatch-benefit model in section 14; both train only on training blocks.

Features include action count, coefficient statistics, constraint structure, circuit family, depth and a cheap incumbent. Exact optima and final evaluation outcomes are offline labels, never hidden runtime inputs. A feature requiring a solver must include that computation in online latency.

Scale and fit budget

For each circuit family, use a deterministic coefficient scale $s_Q = \max(1, \text{the largest absolute nonconstant encoded coefficient})$, in fixed coefficient units, and optimise H/s_Q . Record penalties and s_Q , and use the same transformation at training and inference. Cost angles refer to the scaled Hamiltonian. Undo scaling before comparing physical decision margins. Angle banks remain separate by family and depth.

Use 24 training anchors, C0 and C1, $p = 1$ and $p = 2$, three seeded starts and at most 80 optimiser evaluations per start: at most 23,040 expectation evaluations. C2 adds a separately recorded budget only if its conditional local study is admitted.

Select at most eight donors per family-depth pair on training data. Score them on 120 further training checkpoints and 80 checkpoints from the tuning blocks. Calibration blocks remain untouched until all fitting and tuning choices are fixed. Save bank construction, fit logs, seeds, failed fits and CPU time.

Fit a small inspectable model. Fix optimiser termination and compute budgets; no unrecorded restart replaces a failed fit. Impose angle periodicity only when proven for the implemented gates and weighted Hamiltonian: arbitrary real-weight cost angles need not have period 2π .

Comparisons and interpretation

Compare learned selection with one fixed donor, nearest-neighbour transfer, random donor selection and a per-instance best-found variational reference under a fixed offline fit budget. The reference is not a certified quantum optimum [10-12].

Report optimal-sample probability, raw feasibility, best-of-pool regret and inference time. Define regret as $(f(a) - f^*) / (f_{\max} - f^*)$ on legal actions. If the legal range is zero, every feasible sample has zero regret; an all-infeasible pool still has regret one. Do not use an approximation ratio that becomes unstable under objective shifts.

The proposed non-inferiority margin against the best-found reference is 0.02 in regret. Report an upper one-sided 95% paired block interval for learned-minus-reference regret; crossing 0.02 fails that criterion. This comparison is exploratory in this study. Benefit over fixed and nearest-neighbour transfer needs its own evidence. A null result removes the claimed AI benefit.

12 Circuit families and compilation

Use shallow circuits that can survive the available hardware and budget. More gates are not inherently more novel or useful. The core hardware comparison contains two fixed families, C0 and C1. Select depth on tuning data. C2 belongs to a separate conditional local study and is not included in this hardware budget.

Family	Construction	Role
C0	Transverse-X mixer with penalty cost Hamiltonian	Standard reference
C1	Within-car XY exchanges with one excitation per action block	One-hot-preserving alternative
C2 conditional	Guarded single exchanges and necessary paired exchanges	Test only when genuine joint hard constraints justify it

Use $p = 1$ and $p = 2$ locally. Nominate one depth per hardware arm before calibration and hardware outcomes are visible. Record each arm's parameter count and fitting budget. Comparing a carefully optimised structured circuit with an untuned reference is not acceptable.

For C1, an XY term exchanges occupation between two action qubits and preserves their total excitation. A connected ring of exchanges is a sparse baseline. Test any denser schedule against its routing and native two-qubit-gate cost. One-hot preservation does not imply satisfaction of every joint constraint.

Initialisation and expressivity

Count state preparation in both circuit depth and latency. Use a validated feasible superposition when affordable. If beginning from a single feasible basis state, ensure the schedule first creates an informative superposition: a diagonal cost layer applied to one basis state changes only a global phase. Do not report an ineffective first cost parameter as useful expressivity.

Check mixer connectivity over the enumerated feasible graph on every small instance. A feasible circuit confined to a poor connected component can be inferior to a less constrained sampler. Unitarity, ideal feasibility and connectivity need separate evidence.

Compilation protocol

Pin software versions, native gates, layout policy, transpiler seed and optimisation level. Record logical variables, ancillas, active physical qubits, two-qubit gates, duration, depth, SWAP overhead and calibration timestamps. Include preparation and ancillas in all resource bounds. Use a small fixed layout-search budget identically for the two arms.

Primary hardware data use raw samples, with explicit suppression and twirling settings, no unbudgeted extra executions, and no variational fitting on the QPU. If postselection or repair is explored, report raw feasibility, rejected shots, repair CPU time and the same repair rule for all applicable samplers. A repaired classical output is not a raw hardware success.

QAOA and constraint-preserving mixers are established methods [7-9]. This experiment tests a task-specific use of them.

13 What an advanced circuit must justify

The optional advanced construction is a pit-constraint adaptation of guarded exchange methods, with explicit attribution to prior guarded XY and collective-exchange work [13]. Its research question is whether the extra controls improve useful yield enough to offset compilation and noise.

For two actions i and j within a car's one-hot block, use an exchange only when the unchanged selections of the other car are compatible with both endpoints. The guard must be independent of the occupation being exchanged, so it commutes with that exchange. Verify the implemented unitary against the intended transition matrix, including ancilla uncomputation.

Why connectivity matters

Consider two cars choosing slots A or B, with a declared policy requiring different slots. The legal assignments (A,B) and (B,A) cannot be connected by changing only one car without entering a forbidden state. A paired exchange can bridge them.

This example explains the mechanism; it does not establish that ordinary double stacking is illegal. In the primary physical model, stacking usually has a finite cost. If the real model has no corresponding joint hard constraint, omit the guards instead of manufacturing one.

Enumerate the feasible graph, find connected components under single exchanges, and add only policy-justified paired moves needed to restore reachability. Publish their action semantics and gate decompositions. Do not describe a generic graph bridge as a newly invented quantum primitive.

Admission criteria for C2

The compiler must represent a genuine constraint; ideal circuit checks must show zero infeasible amplitude to numerical tolerance; the feasible graph must have the intended connectivity; and the transpiled circuit must fit the registered local resource bounds.

Development results must then show a useful trade-off against C1 at matched fitting and sampling budgets. Candidate measures are useful feasible probability, best-of-pool loss, native entangling gates and full execution time. A raw feasibility increase alone is insufficient if useful candidates become rarer.

If C2 fails, retain the failure report and close that local substudy. C0 and C1 remain the hardware families. A well-supported conclusion that simpler circuits are preferable is stronger than a circuit made elaborate to satisfy a novelty label.

Any future claim of a new circuit method requires an additional proof or demonstrably distinct construction and a renewed prior-art comparison. This dossier does not make that claim.

14 Margin-aware allocation and acceptance

The allocator starts from a legal state and feasible incumbent. Exact enumeration is mandatory when cheap. If that incumbent attains the exact proxy minimum, record zero proxy headroom and skip further proxy optimisation. Downstream reranking remains a separate task with classical comparators.

A reproducible proposed rule

Define z as the observable state features, current incumbent, remaining effective time and available resource budget. Freeze the generator set and budget grid on tuning blocks. Include a stop action, classical search and the nominated quantum generator. Reject infeasible or over-budget options first.

Using training blocks, fit $g(z, m, b)$, the expected reduction in normalised team loss from generator m with budget b under the complete acceptance-and-fallback policy. Labels use independent simulation banks and modelled latency regimes; they include failed and late outputs. Do not fit this model on tuning or calibration blocks.

After tuning, calibration blocks estimate overprediction. For each block take the maximum positive residual g minus its independently estimated policy benefit over the registered checkpoints and options. Let q be the empirical 95th percentile of these block residuals, conservatively allowing for their Monte Carlo error. The score is $g - q - \lambda u$, where u is estimated QPU seconds. The primary λ is zero; fixed positive values are sensitivity analyses.

Dispatch the highest positive score that fits the time and usage budgets; otherwise stop. Resolve ties in favour of the classical option with lower cost. All inference, feature and incumbent computation consumes decision time. Compare this rule with a fixed threshold and always-classical allocation.

This is an empirical prediction score, not a confidence bound or a proof of conditional improvement. With 24 calibration blocks its tail estimate is uncertain. Report held-out overprediction, harm and coverage; keep shift-panel results separate.

Candidate acceptance

Accept only a timely, current-state-valid candidate that clears the frozen scoring margin under a matched online scoring budget. Check its pit-entry cutoff. Otherwise use the fallback. Do not consult final-score estimates or hidden future draws. Evaluation replays this frozen validation policy causally.

For the separate mathematical bound, let J and f estimate the same outcome in the same units. If $|J(a) - f(a)| \leq \epsilon$ uniformly over all relevant actions, then $J(a) - J(b) \leq f(a) - f(b) + 2\epsilon$. A time-error bound cannot certify rank improvement, and an empirical residual quantile is not a uniform bound.

The allocator must demonstrate value beyond established algorithm selection [27]. Hardware observations remain too few to fit a general cloud-queue model; scenarios must be labelled as assumptions.

15 Deadlines and runtime evidence

Use nominal budgets of 5, 10, 30, 60 and 120 seconds, with 30 seconds primary. These are research settings, not team-supplied requirements. A valid deadline must also respect the physical opportunity to act.

For each checkpoint, record the earliest commitment cutoff for the registered two-car menu and a fixed communication margin. The effective deadline is the earlier of the nominal deadline and that cutoff minus the margin. If no positive time remains, the decision is already closed. Do not move a missed “pit now” instruction to another lap without treating it as a different action.

Time and state advance

Run all arms from the same initial information and commitment protocol. Advance the race under the current feasible plan while computation occurs. At the common commitment epoch validate recommendations against events observed by then; discard stale actions and use the current feasible fallback. Future events remain hidden from solvers.

If the simulator cannot represent this ordering, label the experiment a static checkpoint benchmark. It cannot support an operational pit-decision latency claim. Report effective deadlines and expiry rates alongside nominal budgets.

Full runtime accounting

Record state processing, features, encoding, incumbent search, parameter inference, circuit construction, compilation, submission, queueing, execution, retrieval, decoding, online scoring and validation. Measure the actual critical path when tasks overlap. Offline training is separately reported; state-dependent preprocessing belongs online.

Use monotonic client clocks for durations and UTC timestamps for cross-system audit. Provider clocks require their own origin and uncertainty; do not subtract unrelated clocks without alignment.

At the deadline return the valid incumbent. Late, failed, stale and infeasible outputs remain in the denominator. A late job can still consume QPU time; retain its offline result and charge rather than counting it as a timely recommendation.

Distinguish three latency records

Local measured latency is the executed local system on the recorded machine.

Physical measured latency includes actual client, network, queue and provider timing.

Scenario latency inserts declared delays into local experiments. It is not a measured service-level guarantee.

Do not divide paired-job turnaround by two to estimate standalone latency. The isolated jobs provide only a small separate sample. Test expiry, late retrieval, changed race state, invalid fallback, interrupted persistence and exhausted quota before freezing the protocol.

16 Classical baselines and fairness

Select the operational classical comparator from the included methods using tuning blocks, then freeze it. The test comparison must not use whichever classical method looks weakest or whichever test comparator yields the most favourable result.

Comparator	Purpose and matching requirement
Legal enumeration	Exact small proxy reference; enumerate plans, not invalid bit strings
Independent integer model	Validate formulation and larger-instance bounds
Restricted DP	Strong reference where its state assumptions match
Greedy plus local improvement	Cheap feasible fallback and practical runtime competitor
Uniform legal sampling	Test whether circuit sampling exceeds random search
Simulated annealing or equivalent local stochastic search	Compare classical stochastic candidate generation
Classical simulation-based reranking	Test whether extra computation is better spent evaluating legal plans
Fixed and nearest-neighbour QAOA donors	Isolate the value of learning

Every arm receives the same observable checkpoint, legal action definition and proxy coefficients. Candidate pools differ because generation is the treatment. Use equal effective deadlines for the operational comparison and equal sample or evaluation budgets for mechanism comparisons; report both without conflating them.

Give classical algorithms adequate implementation quality and development tuning. Report thread counts, warm starts, compilation, cache state, hardware and training effort. A Python brute-force loop is not an adequate comparison when a vectorised enumeration or established solver is available.

Offline optimum references may spend more computation, but must be labelled references rather than deadline-feasible methods. Conversely, a reference that actually finishes before the deadline is a legitimate operational competitor.

Required ablations

Remove learned donor selection; replace the structured mixer with C0; remove the uncertainty margin; use always-quantum dispatch in the local or modelled-latency study; use always-classical dispatch; and replace reactive rivals with the frozen non-reactive sensitivity policy. Compare the same final outcome and resource accounting.

Do not remove validation in an operational run to create a “better” unsafe arm. A validator-disabled study is a local diagnostic of failure rates, with no claim of admissible decisions.

No LLM is required in the runtime path. AI-assisted development is disclosed separately from the evaluated learned model.

17 Planned experiment matrix

The matrix separates operational decisions, circuit mechanisms and physical validation. The hardware budget does not execute the whole decision corpus. No planned count is reported as completed.

Panel	Proposed target	Interpretation
Main local decisions	640 checkpoints at the 80-block floor; 10 policy seeds per stochastic arm	6,400 stochastic decisions per arm and nominal budget; blocks remain the independent units
Ideal core circuits	120 cases × 2 families × 2 depths × 30 pool seeds	14,400 pools per parameter policy
Local noisy core	40 cases × 2 families × 2 depths × 10 pool seeds	1,600 pools per parameter policy
Conditional scaling	1,200 generated instances	Requires a separately validated multistop formulation
Hardware quality	12 cases × 3 time blocks × 2 arms	72 pools, with 12 independent cases
Isolated latency	6 cases × 2 arms	12 jobs; descriptive only

Ideal learned, fixed, nearest-neighbour and random-donor policies give 57,600 pools; the variational reference adds 14,400, for 72,000 total. Its fitting cap is another 115,200 expectation evaluations, included in the resource pilot. Noisy learned and tuning-selected simple policies give 3,200 pools. Hardware compares complete frozen circuit-policy packages.

Each mechanism pool targets 1,024 shots. Compute ideal distributions once and resample efficiently; these pools are not additional circuit preparations on physical hardware. Report exact ideal probabilities as well. C2 has its own admitted-case count and local budget; it is absent from these core totals.

Main decision trials compare selected policies on the complete action menu. The 120-case circuit panel may use restricted menus, so it is a separate estimand. Execute deterministic policies once and reuse their records. For stochastic arms average ten policy-seed outcomes within each checkpoint; save all seeds. Different nominal deadlines may require distinct runs because dispatch changes with the budget.

Resource pilot

Use at most 60% of RAM, default noisy cases at 10 qubits or fewer, and checkpoint bounded run units. Estimate CPU-hours, wall time and storage for all fitting and scoring. The provisional 24 CPU-hour ceiling is not evidence that this matrix fits.

Before test access, reduce redundant resampling or amend scope if the campaign cannot fit. Stop at the frozen resource cap and report incomplete units; do not change the design after viewing test effects to obtain a preferred result.

18 Sample size and inference

Shots, policy seeds and simulation worlds are not independent race scenarios. The main synthetic estimand is the equal-weight mean over the eight declared families, their parameter blocks and the balanced SC/VSC checkpoints.

Primary local analysis

Within each checkpoint average the ten stochastic policy-seed outcomes. Within each block average paired classical-minus-proposed loss differences across checkpoints. Give blocks equal weight in the balanced design. Report the mean, a 95% interval, per-block effects and the frequency and magnitude of harm.

Use a stratified block bootstrap, preserving checkpoints, policy seeds and paired-world comparisons as complete records. This primary procedure estimates uncertainty for the realised finite-simulation experiment. Do not automatically add inner world resampling to an outer bootstrap that already includes simulation variation; that can double-count it. Section 19 requires a separate Monte Carlo error diagnostic.

For historical data, cluster by race and report cross-race dependence limits. The synthetic eight-family design does not represent every circuit or the natural frequency of interruptions.

Sizing before test access

Estimate paired block standard deviation s from the reserved calibration blocks after model and comparator freeze. The normal approximation for a two-sided 5% test with 80% power is $n \approx [(1.96 + 0.84)s/\delta]^2$. A 95% half-width h suggests $n \approx (1.96s/h)^2$. Targets are $\delta = 0.02$ and $h = 0.01$.

Use the larger count, account for uncertainty in s , and verify coverage and power by simulation of the planned block analysis. Round up to a multiple of eight, with floor 80 and maximum 160 if feasible. If the requirement exceeds that maximum, describe the study as estimation and report achievable precision.

Illustrative raw counts are 49 for power and 97 for precision when $s = 0.05$; at $s = 0.10$ they are 196 and 385. These are not pilot results. Rounding for family balance follows afterwards.

If cheap exact optimisation makes strictly positive proxy improvement impossible throughout the admitted domain, do not use zero pilot variance to assert that an 80-block superiority study is powered. Apply the headroom gate before opening the test set.

Claims and multiplicity

Only H1 has a planned confirmatory superiority test, conditional on that gate. A 95% interval excluding zero supports a positive average difference; a practically worthwhile gain requires comparison with 0.02. Other deadlines, H2, H3, non-inferiority diagnostics and hardware intervals are exploratory, with every comparison reported.

Report exclusions, missing jobs and deviations. A nonsignificant result does not establish equivalence. No campaign expands after test access merely to reach significance.

19 Monte Carlo design and stopping

Use independent random banks for training labels, online decision scoring and final evaluation. Final-bank seeds, future draws and outcome scores are hidden from fitting and online scoring. During evaluation, frozen policies may observe only events revealed at their simulated time, including causal validity checks at commitment. Selecting a plan using its final-evaluation score would introduce bias.

Common random numbers

Couple comparable exogenous quantities by keys such as block, replication, driver, lap and event type. Assign potential draws even when an action changes whether an event occurs. Merely resetting a global seed is insufficient when different actions consume different numbers of random draws.

Keep action-dependent causal mechanisms in the simulator; coupling does not mean forcing different actions to produce identical traffic or service outcomes. Report the observed covariance of paired losses and compare the variance of differences with independent-draw controls on development cases [14].

Replication counts

On calibration blocks start with 2,048 evaluation worlds per checkpoint. Select a fixed production count from 2,048, 8,192 or 32,768 using paired variance. Target a 95% Monte Carlo half-width no larger than 0.02 at at least 90% of calibration checkpoints. Independently diagnose Monte Carlo error in the primary block-mean effect; the proposed target is half-width no larger than 0.002, one fifth of the primary precision target. Account for ten policy seeds and shared worlds.

If that target cannot be met within resources, report the achieved precision and revise the endpoint or scope before freezing. Do not silently keep simulation running until a preferred sign emerges. Within the test run, use the frozen count and report unresolved comparisons.

For a difference bounded in $[-1,1]$, the worst-case normal-approximation half-width at 8,192 independent worlds is about 0.0217; at 2,048 it is about 0.0433. Actual paired variance may be much smaller. These calculations do not include model error and are not exact finite-sample coverage guarantees.

Use the same final worlds across arms and policy seeds for a checkpoint and evaluate only distinct selected plans when possible. Cache continuations by state, effective commitment time, plan, policies, simulator configuration and random-bank hash. This avoids spending computation repeatedly on identical decisions.

Two uncertainty levels

Show within-checkpoint Monte Carlo error and across-block uncertainty separately. The primary block analysis retains finite-simulation variation. Diagnose that component separately using paired-world variance, respecting shared streams. If it is material, report the finite-simulation estimand or use a pre-frozen variance-components method; do not append naive nested resampling. No interval removes structural simulator bias.

20 The 540-second hardware envelope

Use IBM alone for the core. Let A be the verified free allowance at campaign start. The campaign commitment ceiling is $\min(420, \max(0, A - 120))$ seconds. With the reported $A = 540$ this preserves a 120-second reserve. Recheck availability before each job and never infer paid access from a successful login.

IBM Open Plan uses a rolling 28-day window [21]. Its published 180-minute promotion is conditional on eligibility and opt-in [22]; it is not assumed here. A refreshed or larger allowance does not automatically enlarge this campaign.

Stage	Jobs	Circuits per job	Shots per circuit	Reservation per job
Resource pilots	2	2	512	15 s
Paired quality	36	2	1,024	8 s
Isolated latency	12	1	1,024	7 s
Primary schedule	50	88 pools	88,064 shots	402 s total

This schedule is provisional: $30 + 288 + 84 = 402$ seconds. Its arithmetic is correct, but the individual caps need pilot support. Include all executed PUB broadcasts, reset and repetition delays, suppression settings and provider overhead. Raw shots and QPU usage are different quantities [23-24].

Pilot-based alternative

After the two development pilots and before any main-case outcome, use supported runtime limits and conservative usage estimates to freeze one complete schedule. First consider 512 shots for all matched main comparisons. If overhead still exceeds a cap, reducing repetitions alone does not fix that job.

An explicit alternative is 24 quality jobs at up to 12 seconds each, 12 isolated jobs at up to 8 seconds each and the same 30-second pilot allowance: 414 seconds total. This gives two repetitions of all 12 cases, 38 jobs and 64 pools; at 512 main shots it gives 32,768 shots including pilots. Use it only if every job estimate and the ledger fit. Otherwise stop or issue a pre-main-outcome budget amendment.

Preserve both arms and all 12 instances. After main data begin, do not switch schedules based on observed circuit quality. Budget failure causes a recorded incomplete campaign, not selective replacement cases.

Cases and other providers

Use six SC and six VSC cases, balanced between 6-8 and 10-12 active physical qubits, three per cell, from 12 blocks. Count ancillas and routing qubits; record logical size separately.

Additional providers require verified free physical access and a distinct scientific question. Braket simulator allowances do not provide free QPU jobs [25]; Azure terms require a workspace check [26]. Diversification is not part of the required plan.

21 Hardware execution and interpretation

Proceed only after local correctness, scientific-value and resource gates pass. Freeze the cases, C0/C1 families, depths, parameter policies, layout rules, software, job order and analysis. The two resource pilots may select the registered schedule before main-case outcomes; they do not tune circuit quality.

Use one backend for the primary comparison. Spread the selected two or three blocks across recorded calibration periods where practical. Record actual temporal coverage, rotate requested arm order and save the submission seed. A backend change requires a separately reported stratum.

Accounting and recovery

Use job mode and set a supported maximum execution time on every job. The cap concerns QPU usage, not the decision's wall-clock deadline [23-24].

Let C be charged campaign usage, R unresolved reservations and r the next reservation. Require $C + R + r \leq$ the frozen campaign ceiling. Also require $R + r \leq \max(0, A_{\text{live}} - 120)$, where A_{live} is the currently available free allowance after known charges. This second check does not subtract C again. Count failed jobs and stop when usage is uncertain.

Persist a tagged unique intent and circuit hashes before submission, then the returned job ID immediately. If interrupted before the ID is saved, recover by intent tags. An unresolved submission is not permission to retry. Save raw results durably before reconciliation; charge each provider job once.

These controls cannot guarantee an exact zero-overflow provider timeout. A cap unsupported by the pinned API or inconsistent with estimates blocks submission. A changed allowance never silently permits spending the reserve.

Raw evidence and interpretation

Save samples, total executed shots, job status, usage, client and provider timestamps, native circuits, hashes, backend properties and every decoder result. Preserve late and failed jobs. Repairs never overwrite raw counts.

Compare complete frozen circuit-policy packages. Attribute a difference specifically to the mixer only with matched parameter and preparation ablations. Report raw feasibility, optimal-sample probability, latency and usage for every case and time block.

The 12 cases do not establish broad race superiority or reliability. The 12 isolated jobs provide only six observations per arm, insufficient for accurate queue-tail guarantees. Calibration drift and unfavourable outcomes are reported, not removed. This campaign does not execute the entire 640-checkpoint decision panel on hardware.

22 Conditional readiness analysis

Define $p\tau$ as the probability that one raw sample is feasible and beats its stated incumbent by a frozen proxy margin τ . Component tests use the frozen greedy incumbent; operational tests use the actual classical incumbent. Also report exact-optimum probability within a pre-frozen numerical tolerance. Proxy improvement is not necessarily race-outcome improvement.

Under an independent identical-shot model, the probability of at least one such sample in S shots is $1 - (1 - p\tau)^S$. To reach success probability at least $1 - \alpha$, the required sample count is the ceiling of $\log(\alpha) / \log(1 - p\tau)$, for $0 < p\tau < 1$.

For example, $p\tau = 0.01$ needs 299 shots for 95% success, and $p\tau = 0.001$ needs 2,995. These are conditional calculations. For an estimated $p\tau$, propagate a lower interval bound into the requirement. Zero observed successes do not prove $p\tau = 0$; a zero lower bound supplies no finite certified shot requirement. Only an exact objective argument establishes impossible improvement. Drift can invalidate the independent-shot model.

A full boundary

The quantum path must fit state processing, encoding, compilation, queueing, execution, retrieval and validation inside the decision deadline, while providing enough useful probability to justify replacing the incumbent.

Report a surface over queue delay, circuit duration, shot count and useful yield. Mark measured points separately from noise simulations and hypothetical scenarios. A single average fidelity is insufficient: error location, connectivity, parameter quality and the feasible subspace also matter.

If the incumbent already attains the exact objective optimum, $p\tau$ is zero for any strictly positive improvement margin. No finite shot count creates missing headroom. This is an important boundary, not a failed plotting case to remove.

Future hardware claims

A scenario with a shorter queue or fewer gate errors shows what would be required under its assumptions. It does not prove that a provider will deliver those conditions, that the algorithm retains its quality at larger size, or that classical methods will remain unchanged.

Do not translate a small NISQ noise sweep into fault-tolerant logical-qubit requirements without a separate algorithm, error-correction and resource-estimation analysis. Do not fit an exponential quantum speedup from small problem sizes.

Practical output

Produce a rerunnable table stating, for each instance family, the strongest classical result, available headroom, acceptable model error, useful-sample requirement and full latency allowance. A team can use this to reject unsuitable integrations or identify a narrow future shadow-test condition. That is a concrete decision-support output even where quantum is not selected.

23 Cursor and the one-command workflow

Implementation happens after this dossier is finalised. ChatGPT supplies complete prompts with scope, files, acceptance checks and expected evidence. Cursor performs the coding and local execution. The user should not need to invent commands, edit circuits or repair scientific logic manually.

Project isolation

Use a new repository where possible. If a shared repository is necessary, create a separate top-level research directory and allow writes only there. Record hashes of the closed study's protected files before and after a run. A changed protected file is a hard failure; never silently “clean up” the earlier project.

Meaning of do a run

The command resolves the next authorised unit of the frozen run plan, reports its identity and executes it once. Initially, this is local only. Hardware mode is unavailable until the user has explicitly authorised an identified IBM block after reviewing its budget and frozen matrix.

Once that hardware authorisation is recorded, “do a run” may execute the next unit within its remaining scope and cap. It never expands authorisation to a paid plan, another provider, a larger shot count or a new experiment.

Suggested internal states are DRAFT, LOCAL_READY, LOCAL_COMPLETE, PROTOCOL_FROZEN, HARDWARE_READY, RUNNING, RECONCILE_REQUIRED and COMPLETE. A stale protocol hash, missing data permission, unresolved job or inadequate quota prevents transition to hardware.

Each run must finish with a receipt

Report run ID, protocol and code hashes, planned and completed units, actual failures, raw evidence paths, summary paths, elapsed time, actual QPU usage, reserved usage and the next permitted unit. If a process stops midway, resume completed work by manifest rather than repeating it.

GitHub is a separate action

“Push to GitHub” means review the scoped diff, exclude secrets and restricted data, verify tracked manifests and push only the requested project changes. The run command saves locally by default. It does not automatically publish or modify the closed paper.

Do not commit credentials, account tokens, raw protected timing feeds, unverifiable result tables or generated claims with no evidence row. Provider credentials remain in an approved local secret mechanism.

The orchestration layer can be deterministic. The runtime AI claim rests on the evaluated learned policy. Use “agentic” only if an implemented component actually observes, selects actions and manages a bounded goal; do not relabel ordinary helper functions as autonomous agents.

24 Evidence schema and research integrity

Store immutable raw evidence and regenerate derived summaries from it. Every plotted value must be traceable to a run, configuration, raw record and analysis version.

Record	Minimum required fields
Protocol	Version, hypothesis, endpoints, splits, counts, seeds, exclusions, budget and hashes
Checkpoint	Block, family, split, state, units, availability, commitment cutoff and assumptions
Instance	Action dictionary, constraints, coefficients, scale, penalty proof, exact reference
Learned policy	Training IDs, features, donor bank, model, fit seeds, validation decisions
Circuit	Family, depth, angles, initialisation, layout, native gates, duration, hashes
Job	Intent token, provider ID, status, caps, usage, timestamps, backend snapshot
Decision	Policy seed, incumbent, candidates, score, validity, expiry and commitment time
Evaluation	Independent bank, simulator version, rival policy, outcomes, uncertainty
Deviation	What changed, why, when, test exposure and effect on interpretation

Use one record per failed attempt as well as one per success. A run ledger links requested units to completed units; a count in a report is computed from that ledger. Numeric outputs never originate in manually written narrative.

Independent correctness checks

Check QUBO energies against a direct action-cost function; solver answers against enumeration; circuit probabilities against independently computed small matrices; checkpoint resumption against uninterrupted simulation; and derived reports against raw counts. These tests address concrete risks, not superficial coverage percentages.

Claims ledger

For each intended claim record supporting runs, analysis, uncertainty, limits and citation. Mark claims unsupported until evidence exists. Distinguish simulated, noisy-simulated and physical results in every figure legend and table.

Generated code and prose require human responsibility and verification. Disclose ChatGPT and Cursor assistance accurately, including implementation, analysis or drafting if used. Do not claim manual coding or human-only authorship of work actually generated with assistance.

The previous study may be cited for genuine architectural lineage once its final public reference is confirmed. Do not invent a DOI, publication status or peer review. Its observations are not independent replications of this new experiment.

No evidence file may be edited to align with an expected conclusion. Corrections create new versions with an explanation and preserve the original record.

25 Implementation sequence

Each Cursor prompt should implement one reviewable stage and return an evidence receipt. The sequence is intentionally ordered so that expensive or irreversible work cannot conceal defects discovered earlier.

Stage	Work	Exit evidence
1	Isolated project, environment detection, dependencies and run ledger	Clean install, protected-path check, local dry run
2	Scenario generator and causal checkpoint schema	Reproducible states, provenance, split and rejection audits
3	Simulator adapter and independent checks	Resume equivalence, mechanism tests, analytical comparisons
4	Action model, QUBO and independent classical references	Exact agreement on small legal instances and penalty checks
5	Parameter bank, learned selector and circuit families	Split integrity, ideal-circuit checks, training receipts
6	Local pilot, precision and resource estimates	Frozen counts, machine capacity, novelty comparison and deviations
7	Full local study and primary analysis	Complete manifests, intervals, ablations and headroom report
8	Hardware preflight and explicitly authorised block	Balance, free plan, budgets, pinned matrix and resumable intent
9	Hardware analysis and conditional boundaries	Raw job evidence, usage reconciliation and bounded conclusions
10	Reproducibility release and later article	Clean-room reproduction, claims ledger and reviewed outputs

The first coding prompt must not request the whole project plus hardware execution in one uncontrolled step. It should establish the environment and evidence system, then stop with a concrete report for the next prompt.

At stages 3-6, a defect can trigger redesign before the test split is opened. At stage 7 or later, a substantive methodological change creates a protocol amendment and a clearly separated exploratory analysis or new test set.

Local environment choices

Prefer a supported Python environment with NumPy, SciPy, a free integer solver, a small machine-learning library and pinned Qiskit/Aer where compatible. Inspect the actual machine before choosing versions or parallelism. A legacy simulator dependency is isolated rather than forcing the whole environment onto an unsupported stack.

26 Gates and useful outcomes

Gate A, provenance. Every required software and data dependency has a recorded licence or permission. Generated inputs remain the default. Failure blocks the affected input, not unrelated local work.

Gate B, decision validity. The checkpoint, continuation, SC/VSC behaviour and team interaction mechanisms pass independent checks. Failure narrows the domain or requires repair before comparisons.

Gate C, formulation. Direct costs, QUBO energies and independent classical references agree on small cases. The legal action count and compiler cost are reported. Failure blocks quantum experiments.

Gate D, learning and circuits. Models respect separate training, tuning and calibration blocks. Core C0/C1 circuits pass preparation, transition and compilation checks. C2 is separately gated. Held-out claims must use the frozen comparisons.

Gate E, scientific value. Before the full test campaign, record whether exact classical solution removes all proxy headroom. If so, omit an impossible superiority test. Proceed only with a separately justified boundary or mechanism question, or validate a larger model before freeze. If no substantive contribution survives comparison with prior work, redesign before QPU use.

Gate F, precision and budget. The chosen experiment count follows development-only variance and resource evidence. Hardware reservations fit verified free access with the reserve intact. Failure leads to a smaller explicitly limited study or a later free allocation, not unapproved spending.

Outcomes that would help a strategy group

A validated classical-first fallback and causal checkpoint interface can improve reproducibility of strategy research. A measured map of proxy error can show where better modelling matters more than faster optimisation. A candidate-yield and latency boundary can prevent premature quantum integration. A successful learned circuit policy could justify a carefully scoped follow-up.

These are proposed benefits. Demonstrated lap-time, points or operational gains for a real team require appropriate real-world evaluation. A synthetic benchmark must not be marketed as evidence that an actual race result would have improved.

Career evidence

Prepare a concise technical portfolio after results: an executable benchmark, a transparent hardware record, a reproducible negative-and-positive findings report, and a short explanation of the integration conditions. Explain the author's scientific decisions and ability to audit AI-generated code.

This can provide credible evidence of skill across AI, quantum methods and motorsport decision-making. It cannot make the author a guaranteed recruitment target. The strongest positioning comes from solving and explaining a specific problem well.

27 Publication and claim discipline

The later article should follow the completed evidence: motivation, related work, causal decision model, algorithm and circuit definitions, protocol, results, robustness, limitations and reproducibility. Journal selection follows the demonstrated contribution, available validation and verified fee policy.

Do not assume a particular journal accepts synthetic-only evidence, a negative hardware study or a DBA framing. Do not assume an article-processing charge will be waived. The zero additional spending requirement applies to publication choices too.

Allowed only with matching evidence

“The learned donor selector improved useful probability on the held-out generated panel” requires that exact comparison and interval. “The structured circuit reduced invalid raw samples on IBM” requires physical paired evidence. “The allocator improved simulated team outcomes” requires the independent evaluator and full deadline accounting.

“The system improved Formula 1 performance,” “quantum advantage,” “first quantum F1 optimiser,” “guaranteed future superiority,” “real-time team deployment,” and “a new guarded mixer” are not supported by this plan.

Important remaining limitations

Synthetic parameter blocks are not actual races. Historical counterfactuals remain model-based even with lawful data. Model assumptions can favour a solver. Small circuits may not represent large problems. The physical sample is small and backend-specific. Queue conditions and provider policies change. A locally measured CPU budget is not a team's production environment.

General novelty cannot be established by a finite literature search. A claim of originality must identify the precise algorithmic or empirical difference from the closest work and show why that difference is useful.

What would justify proceeding to a manuscript

The implementation must be reproducible; the main scientific claim, whether positive or negative, must be supported by the planned comparisons; uncertainty must match the actual sample; and the contribution must remain meaningful after limitations are stated plainly. A successful pipeline run by itself does not meet those conditions.

If the results reveal no useful quantum role, report the boundary and the reasons. If they reveal no meaningful novel contribution at all, do not manufacture one with stronger language. Use the evidence to define the next research question.

The aim is a defensible contribution with a credible path toward real motorsport validation, not a predetermined positive story about quantum computing.

28 Research references

These references were checked on 18 September 2026. Preprints are identified as preprints. Sources establish prior work or implementation facts; proposed design choices in this dossier are not attributed as established results.

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Search and verification limits

The review covers race optimisation, QUBO, learned strategies, counterfactuals, parameter transfer, mixers, algorithm selection and resource policies. It does not prove absolute novelty. Before freeze, compare precise claims with the nearest methods and record unavailable content. Recheck bibliographic metadata and interfaces before publication. Unselected commits, package versions, account eligibility and simulation parameters remain implementation tasks, not verified facts.

30 Protocol freeze record

This dossier is a design recommendation. The following fields become concrete during the implementation and development pilot. No test outcome may be used to choose them.

Field	Required freeze evidence
Scope and rights	Core generated domain; any permitted historical extension and licence
Simulator	Code hash, checkpoint semantics, mechanism tests and parameter distributions
Corpus	Family definitions, separate fitting/tuning/calibration/test blocks, counts and seeds
Objective	Units, risk weights, legality, penalty derivation, normalisation and tie rules
Algorithms	Classical comparator, model features, donors, circuits, depths and tuning budgets
Statistics	Headroom gate, estimand, policy seeds, effect margin, sizing and resampling rule
Simulation	Random-stream design, production replication count and final-bank isolation
Deadlines	Clock origin, action cutoffs, effective budgets, state advance and fallback validity
Hardware	Verified free access, available balance, reservation ledger and frozen case matrix
Execution	Authorised run scope, resume rules, protected paths and evidence schema
Claims	Prior-art comparison, intended claims, unsupported claims and known limitations

Completion standard

Before declaring a local campaign complete, reconcile planned versus actual cases and repetitions, check all evidence files, reproduce tables from the raw records and report missing units.

Before declaring hardware complete, reconcile every intent, job status and usage entry, preserve all returned data and account for the reserve. A target count does not override the budget.

Before drafting the article, confirm that its central claim survives the strongest fair comparator, has adequate stated precision and is supported by the level of domain validation actually achieved.

Recommended next action

Use this version as the basis for the first scoped Cursor setup prompt. Begin with project isolation, environment inspection, the run ledger and generated-data schema. No quantum execution is needed for that stage.